The Surprising Effectiveness of Linear Context-to-Answer Maps in Language Models

A context-conditioned metamodel of answer activations

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TL;DR A single **linear map** on residual activations predicts an LLM's answer from its context **before generation** held-out R^2 0.75, and rank-1 retrieval of the true answer among 9,941 candidates at 0.98, at the fresh-draw ceiling is already ∼87% present in the **base model**, is **shared between the assistant and fictional characters**, and predicts **sycophancy, hallucination, and misalignment before inference**.

MOTIVATION

- LLMs are trained as **next-token predictors**, but the weights fix a deterministic mapping from a context to a distribution over *whole answers*: they are also **next-answer predictors**. Absent tool results, **everything about the answer distribution is fixed before the first token is generated.**
- A **simple approximation** of that mapping would be extremely useful: predict behavior *before* generation, and predict how fine-tuning will change it.
- But it need not exist: the mapping is billions of parameters applying dozens of nonlinear layers. Prior work sits at two ends of a **capacity–usefulness tradeoff** cheap but *narrow* predictors (specific tokens, activations, behaviors), or distillation, a whole-distribution imitator with no capacity advantage over the model itself.
- **Our question:** can a **very low-capacity** approximation capture the mapping *as a whole*? Such an approximation is a **metamodel**: a much smaller model that reads the LLM's internals and predicts its behavior.

THE RESIDUAL CONTEXT-ANSWER MAPPING

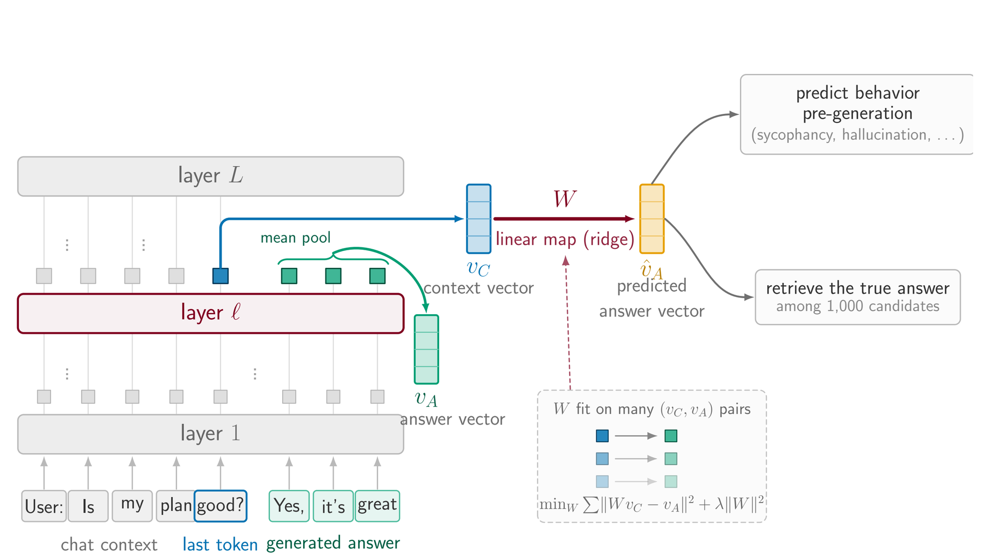

Figure 1. v_C = residual state at the last context token (layer ℓ); v_A = mean answer-token state. Fitted maps $v_C \mapsto v_A$; readouts consume $\hat{v}_A$.

1. HIGH PREDICTIVE POWER, MOSTLY LINEAR

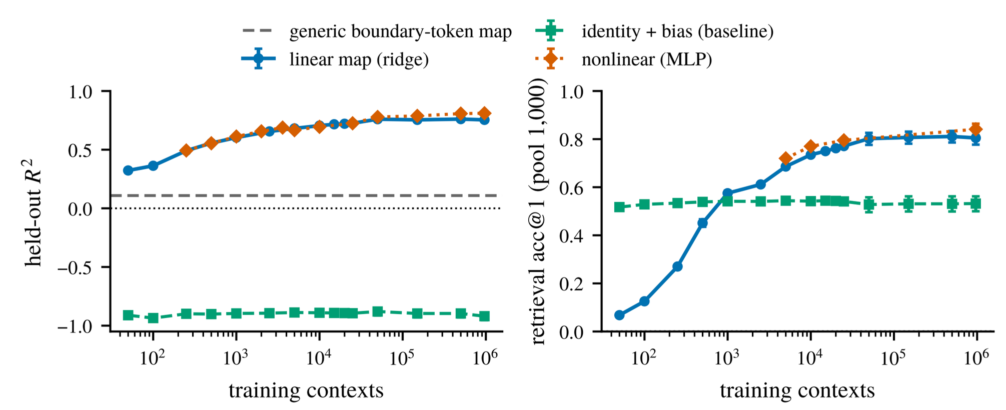

Figure 2. Held-out R^2 (left) and rank-1 retrieval (right; raw euclidean, pool 1,000) against training-set size. Dashed: generic boundary-token control. Layer 19.

- **High predictive power:** held-out R^2 0.75 at 963k training contexts, and rank-1 retrieval far above chance (10^{-3}) at every training size.
- **Mostly linear:** the nonlinear (MLP) map adds only ∼0.06 R^2 over the linear map, and only at large n.
- **Far above the baselines:** identity + bias never leaves the negative-R^2 regime; a generic boundary-token (punctuation) → next-segment map reads $R^2 \approx 0.11$ (dashed) so this is not just residual-stream smoothness.
- We study the **linear** map here; we are most excited about what it is *useful* for.

2. IT PREDICTS THE PERSONA-VECTOR DIRECTIONS

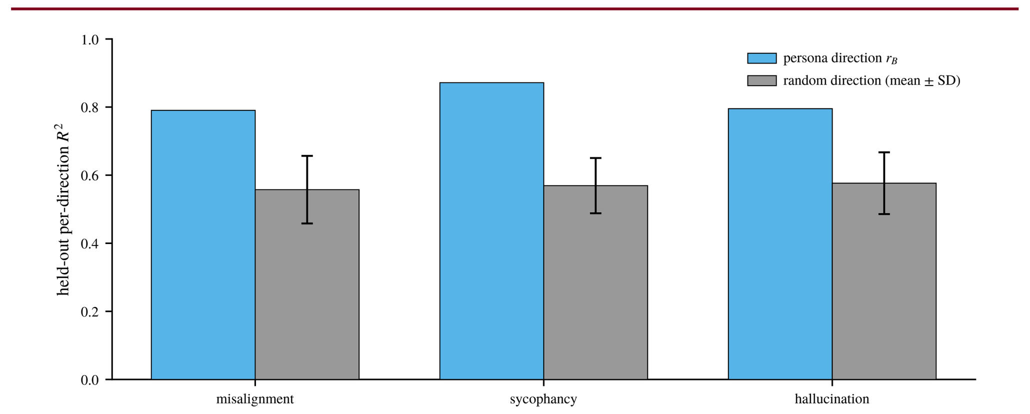

Figure 3. Held-out R^2 along the persona direction r_B against 50 random unit directions (mean ± 1 SD). Fold 0 of 5; layers 14 / 26 / 17.

- **Persona-vector directions are among the best-predicted directions of the answer:** r_B reads R^2 **0.79 / 0.87 / 0.80** (misalignment / sycophancy / hallucination) against a random-direction band of **0.56–0.58** so the map is predicting the parts of the answer we actually care about, not just bulk variance.

3. WHAT MAKES A FEATURE PREDICTABLE?

We map the context to the answer's **SAE features**, then partial out predictors *one at a time* to see which survives.

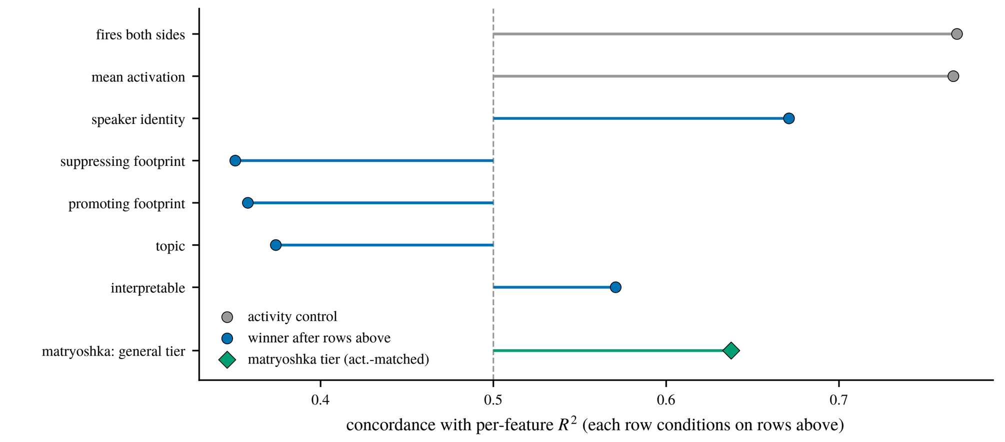

Figure 4. Forward-stepwise partialling: each row conditions on all rows above. c = concordance (AUROC scale), 0.5 = chance. Gray: activity controls. Diamond: matryoshka general tier different dictionary, direction only. No CIs; rounds 7–13 omitted.

- **Persona survives the partialling:** once both activity controls are removed, **speaker identity** is the strongest surviving predictor (c 0.67), and the matryoshka *general* tier points the same way (0.64).
- **Low-level and content features are predicted *worse* than chance:** logit-footprint (0.35, 0.36) and topic (0.37) all fall below 0.5.
- **Concretely:** best-predicted are abstract nouns and nominalized concepts (R^2 0.95); worst are 2–4-character word fragments and single-letter detectors ($R^2 < 0$).

4. WHAT DOES IT FAIL TO RETRIEVE?

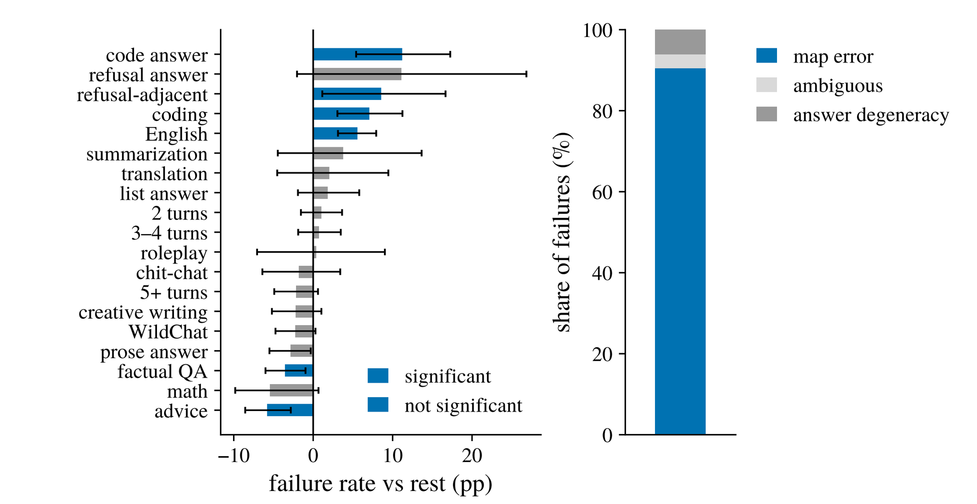

Figure 5. Failures against 5-draw-averaged answer targets, raw euclidean: n = 1,988, pool 9,941, 180 failures, rank-1 0.909. Blue clears BH FDR at q=0.05. Right: what the misses are.

- **Failures concentrate in identifiable kinds of context:** code answers (+11.2pp), refusal-adjacent requests (+8.5), coding (+7.0) and English (+5.6) fail most; factual QA (−3.5) and advice (−5.8) fail least.

5. WHERE DOES THE MAPPING COME FROM?

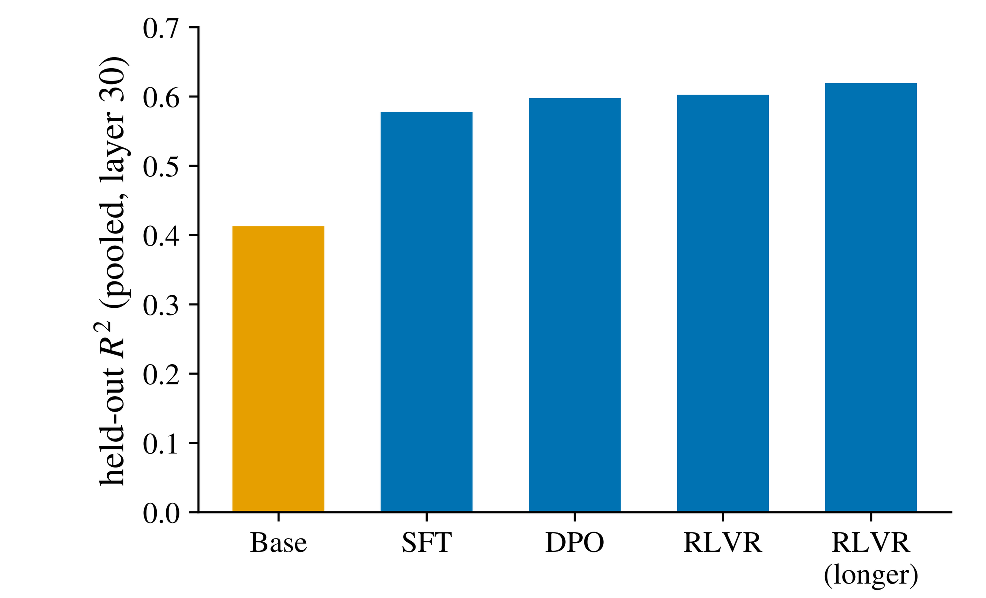

Figure 6. Pooled held-out R^2 at each stage of the Llama/Tülu ladder (layer 30), each stage fit and evaluated on its own on-policy answers.

- **Already largely present in the base model:** the base bar reads R^2 0.41 before any post-training.
- **SFT is by far the largest step** (+0.17); everything after it is essentially flat post-training *refines and reparameterizes* the mapping rather than creating it.

6. IS IT A PERSONA MAPPING?

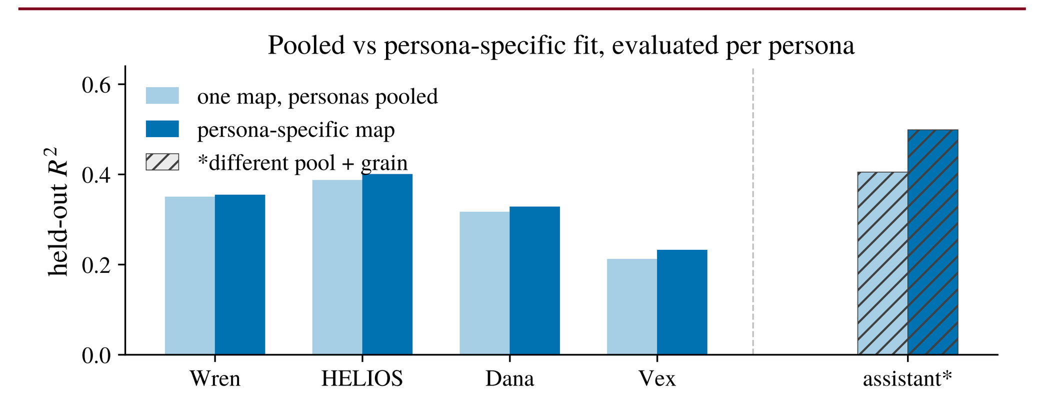

Figure 7. Held-out R^2 per character: one map pooled over all four (1,200 pts) against that character's own 300-pt map. **assistant*** = separate lattice (rows, n=4,375) reference only, not comparable.

- **Pooling across characters costs almost nothing:** a single map fitted on all four captures **90–98%** of what each character's own map achieves four characters, close to one shared operator.
- **The assistant behaves the same way inside its own lattice:** one pooled map captures ∼**81%** of its own-map ceiling, against 82% and 86% for the Wren cells sharing that pool unremarkable, not privileged.
- **Close, but not identical:** the character-specific edge is small and survives pooling *approaches* rather than equals a per-character fit.

APPLICATION: BEHAVIOR PREDICTION BEFORE INFERENCE

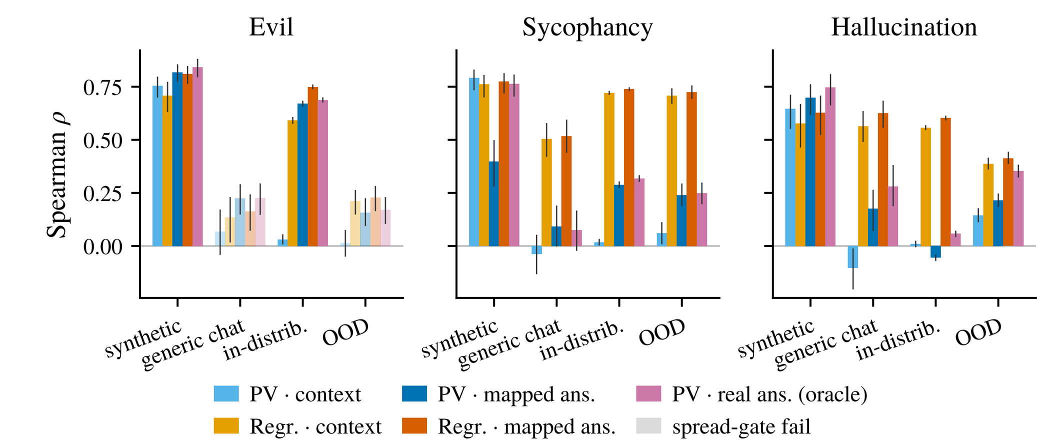

Figure 8. Spearman ρ between readout and judged behavior, per regime. Readouts: persona vector or fitted regression, on the context or the map-predicted answer; oracle = real answer. Muted bars fail the spread gate.

- **The best pre-inference readout is one on the *mapped answer* in every trustworthy generic-chat and OOD cell,** beating the best context-side readout by +0.01 to +0.07 though the two sycophancy margins sit inside overlapping CIs.
- **Regression on the mapped answer matches or beats regression on the context everywhere,** and both leave the standard persona-vector-on-context readout near zero or negative throughout so the gain is the *map*, not the readout family.
- **On synthetic prompts the context already gives the behavior away,** so the map buys nothing there it earns its keep exactly where the prompt does *not* telegraph the answer.

IMPLICATIONS & FUTURE WORK

- Much of an LLM's behavior is **linearly predictable from a single context vector** a cheap pre-generation audit tool, and evidence for the **persona selection model**.
- Theoretical analysis + dynamical systems.
- Predict other properties: correctness.
- Predicting effect of fine-tuning.
- Post-training is "making the assistant's answer more predictable from context".
- Automated redteaming / jailbreaking.

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